Chemical equations

Learning outcomes

By the end of this section you should be able to deduce chemical equations, including state symbols, when reactants and products are specified.

The chemistry of cooking

A pancake is a flat, circular cake made from flour, eggs and milk and cooked on a griddle or frying pan. You may have tried other similar versions of pancakes from around the world such as dosa, crêpe, blini and okonomiyaki; all of which are the same shape and made with similar ingredients (**Figure 1**).

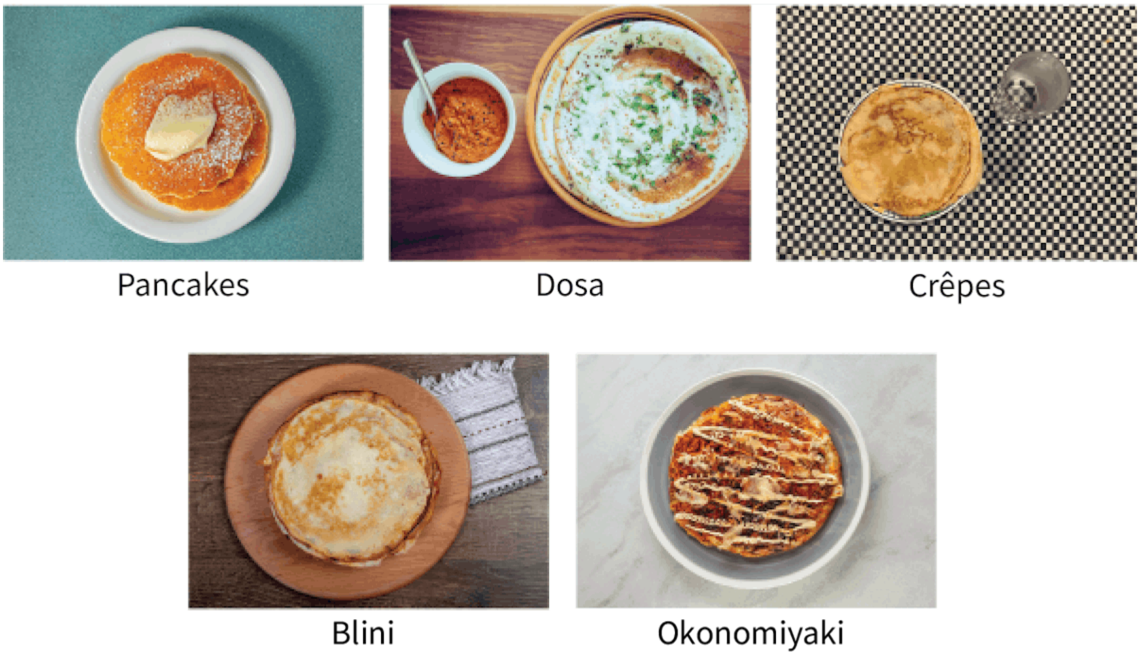

Figure 1. Pancakes, dosa, crêpe, blini and okonomiyaki.

Credits: Chris Schneider / 500px, Getty Images, GCShutter, Getty Images, Andres Jabois, Getty Images,

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The ingredients to make two western-style pancakes are one cup of flour, two small eggs and one cup of milk. This list of ingredients could be rewritten in a similar format to how a chemical reaction is represented:

$$1 \text{ cup flour} + 2 \text{ small eggs} + 1 \text{ cup milk} \rightarrow 2 \text{ pancakes}$$

Notice how the quantity of each ingredient is represented by using a number placed before it. For convenience, a chef could also choose to abbreviate the ingredients and use a letter to represent each food:

$$1F + 2E + 1M \rightarrow 2P$$

What are the benefits of choosing to represent a recipe in this way?

Chemical equations

Chemical reactions are written similarly to the pancake recipe. The names of the reactants and products make up the word equation, but the chemical equation is written using the symbols from the periodic table which make up their chemical formula.

Study skills

When writing the symbols of elements from the periodic table be careful to make a clear distinction between the upper-case and lower-case letters. If you write them incorrectly then you will lose the marks in an exam — CO means something quite different to Co.

Making connections

Converting the names of the reactants and products into their chemical formulas is often the part which students struggle with the most. You may find it helpful to recap how to do this before you begin applying the steps above:

- naming ionic compounds (see subtopic S2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/)).
- naming molecular compounds (see subtopic S2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/)).

- naming organic molecules (see subtopic S3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/)).

Reactants are always written on the left of the arrow and products on the right side. The arrow is important in the equation as it represents a boundary between the reactants and products and also indicates the direction in which the reaction is proceeding. For example, a reaction between potassium metal and water (the reactants) will produce potassium hydroxide and hydrogen gas (the products):

$$2K + 2H_2O \rightarrow 2KOH + H_2$$

In the equation above you can see that each of the reactants and products are represented using their chemical formulas. Each reactant or product also has a number before it to represent the quantity needed.

$$2K + 2H_2O \rightarrow 2KOH + H_2$$

In order to make sure that the correct amounts of each reactant and product are present it is important to make sure that the equation is balanced so that the equation observes the law of conservation of mass. You will likely have learned how to balance equations in your previous studies.

Study skills

Make sure to always use an arrow when separating the reactants and products (definitely not another symbol such as an equals sign).

You may encounter another variation of this arrow in section R 2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/), known as a ‘double arrow’ ($\rightleftharpoons$), which is used to indicate an equilibrium reaction.

You may also recall other symbols being included above the arrow, such as the words ‘pressure’ or ‘heat’. You may also see the symbol ‘ΔH’ (to indicate that energy has been transferred), the chemical formula for a catalyst that has been added or the word ‘electricity’, to indicate that electrolysis is taking place.

Worked example 1

Balance this equation using the simplest whole numbers:

$NH_3 + O_2 \rightarrow N_2 + H_2O$

Step 1: Count the number of each element on both si	$NH_3 + O_2 \rightarrow N_2 + H_2O$

	Reactants	
N	1	
H	3	
O	2	

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des.

Step 2: Find the lowest common multiple for each elemen

$$2NH_3 + O_2 \rightarrow N_2 + H_2O$$

	Reactants	
N	2	
H	3	
O	2	

t, for example, the lowest common multiple for nitrogen and is 2 ni

trogen. Alter the coefficients to achieve this.	
Step 3: Contin	Remember - do not alter the subscripts. $$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$

	Reactants	
N	4	
H	12	

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ue to do this until all elements are present in equal amounts in t

	Reactants	
O	6	

Kog

he reactants and products.	

You should have the same number of each element on both sides of the arrow. Remember that you can adjust the coefficients in front of the reactants and products to achieve this, but you cannot change the subscripts. Recall the pancake recipe we previously discussed – when we wanted to specify that the pancake recipe has two eggs in it we wrote it as 2E with the coefficient placed before the eggs. It would be incorrect to write it as E_2 by changing its subscript, as this would indicate that the 2E's are chemically bonded together.

Study skills

As part of subtopic R2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/) you will have encountered equations which have been balanced using fractions rather than whole numbers. This is particularly important for certain enthalpy changes such as the enthalpy of formation, where the equation must be written to form one mole of a substance. For example, the thermochemical equation used to represent the enthalpy of formation of ethanol is most correctly written as:

$$2C + 3H_2 + \frac{1}{2}O_2 \rightarrow C_2H_5OH$$

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It is possible to balance all equations using fractions, but assessment questions will usually have whole numbers so that's what you should practise using.

There are some key features absent from each of the equations above. Consider the pancake recipe we have been discussing.

Option 1: When selecting the ingredients – eggs are in the liquid state when raw and become a solid when cooked. If you have never cooked before, you might be unsure whether this recipe requires raw or cooked eggs.

Option 2: Will the pancakes produced be a solid or a liquid? Providing information about the physical state of the product will make it easier for the chef to understand how long to cook it for and how best to transfer the pancake from the saucepan to its serving plate.

This is also important in chemical reactions. Specifying the physical state of the species involved is relevant as it will determine other features of the reaction, such as how quickly it occurs.

State symbols

Recall your knowledge of the phases of matter from section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/). Chemical equations will typically also include the physical state of the reactants and products in equations using state symbols.

For example, the boiling of water could be simply represented by

$$H_2O(l) \rightarrow H_2O(g)$$

This is an example using a chemical equation to represent a change in the state of matter, but all other reactions will include them too. The state symbols commonly used are (s) for solid, (l) for liquid, (g) for gas, and (aq) for aqueous, meaning the substance is dissolved in water.

You can see an example of this in **Figure 2**.

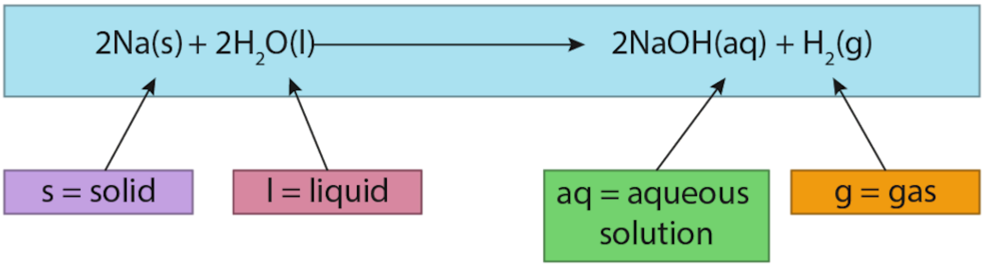

Figure 2. State symbols for a balanced chemical equation.

More information for figure 2

The image displays a balanced chemical equation: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$. This equation represents the reaction between sodium and water. Each component of the equation is annotated with state symbols: 's' denotes solid, 'l' depicts liquid, 'aq' stands for aqueous solution, and 'g' indicates gas. These annotations are color-coded and have lines pointing to the respective parts of the chemical equation to clarify the physical state of each reactant and product.

[Generated by AI]

Try to add state symbols to the potassium and water reaction. You may recall from section S2.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/) that ionic compounds are soluble in water.

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

Why do you think the state symbols are important? This is because they can affect the outcome of a chemical reaction. For example, the rate of a reaction may differ depending on whether the reactants are in the solid, liquid, or gaseous state.

Also, some reactions can only occur in certain states, such as reactions that occur in aqueous solutions. Therefore, including state symbols in a chemical equation helps to provide a more accurate description of the reaction taking place.

Although there can be exceptions to any 'rules' we have to assign state of matter, you will find the **Table 1** useful in helping you determine the state of matter for each component in the equation.

Table 1. The states of matter of some common elements and

Chemical species	Example	Physical state
Metals	Sodium, Na Magnesium, Mg	Solid
Non-metals	Hydrogen, H_2 Oxygen, O_2 Chlorine, Cl_2 Fluorine, F_2 Noble gases	Gases
Ionic compounds	Sodium chloride, NaCl Calcium carbonate, $CaCO_3$	Solids at room temperature Many dissolve in water, forming ac solutions
Molecular compounds	Carbon dioxide, CO_2 Water, H_2O Sucrose, $C_{12}H_{22}O_{11}$	May be solid, liquid or gas dependin intermolecular forces which occur b molecules

Writing a chemical equation

A chemical reaction may be described using a short paragraph. The equation is simply the most universal method of representing this same information concisely. There are a number of steps to follow to help you convert the information into a chemical equation but remember that practice is key.

Steps to write a chemical equation:

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1. Write down the word equation, placing reactants on the left of the arrow and products on the right of the arrow.
2. Convert the names of each of the reactants and products into their chemical formulas.
3. Balance the equation.
4. Add state symbols.

Making connections

As part of topic R3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/) you will learn how to write half-equations. These show the movement of electrons in oxidation-reduction reactions and look quite different to a regular balanced chemical equation.

When might it be useful to include half-equations?

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate engagement with issues of global significance
- Demonstrate how to initiate and plan a CAS experience

Chemical reactions have led to significant improvements in the quality of life for many people around the world. One such example is the Haber process (https://ourworldindata.org/how-many-people-does-synthetic-fertilizer-feed) that produces fertiliser which is then used to grow crops. However, these improvements have not been evenly distributed throughout the world and have led to inequalities in areas such as food security (https://www.who.int/news/item/06-07-2022-un-report--global-hunger-numbers-rose-to-as-many-as-828-million-in-2021).

What can you do as an IB student to help those less fortunate? Possible CAS experiences could be to organise a food drive (https://www.feedingamerica.org/ways-to-give/food-drives) or to support a charity in your local community.

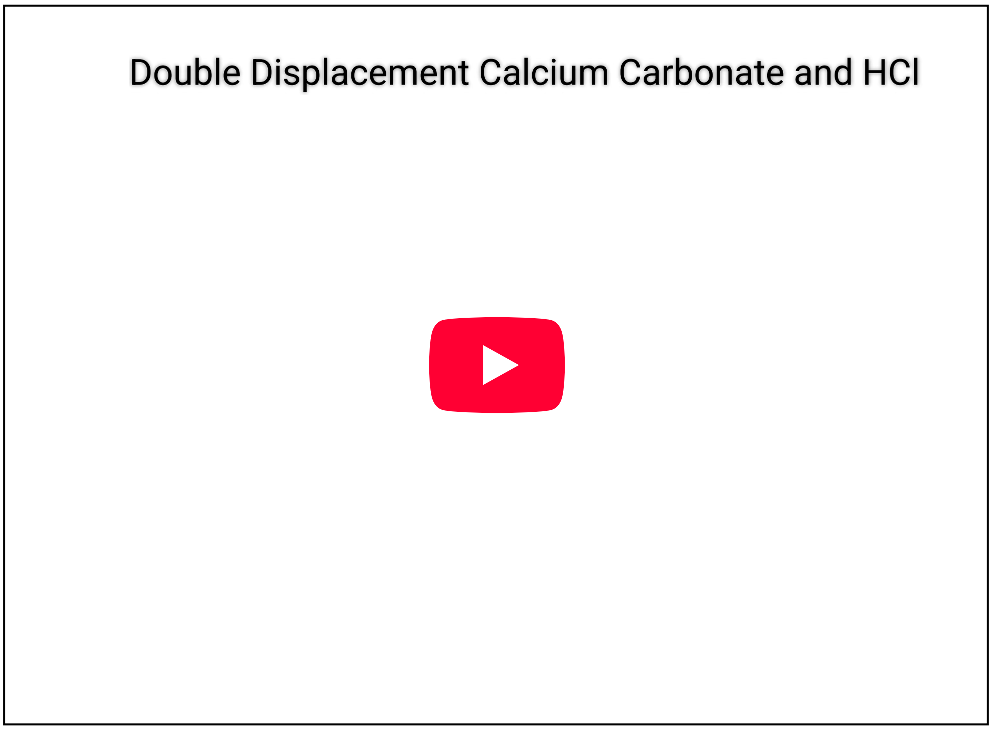
The next activity will allow you a chance to practise converting different descriptions and videos of reactions into balanced chemical equations.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

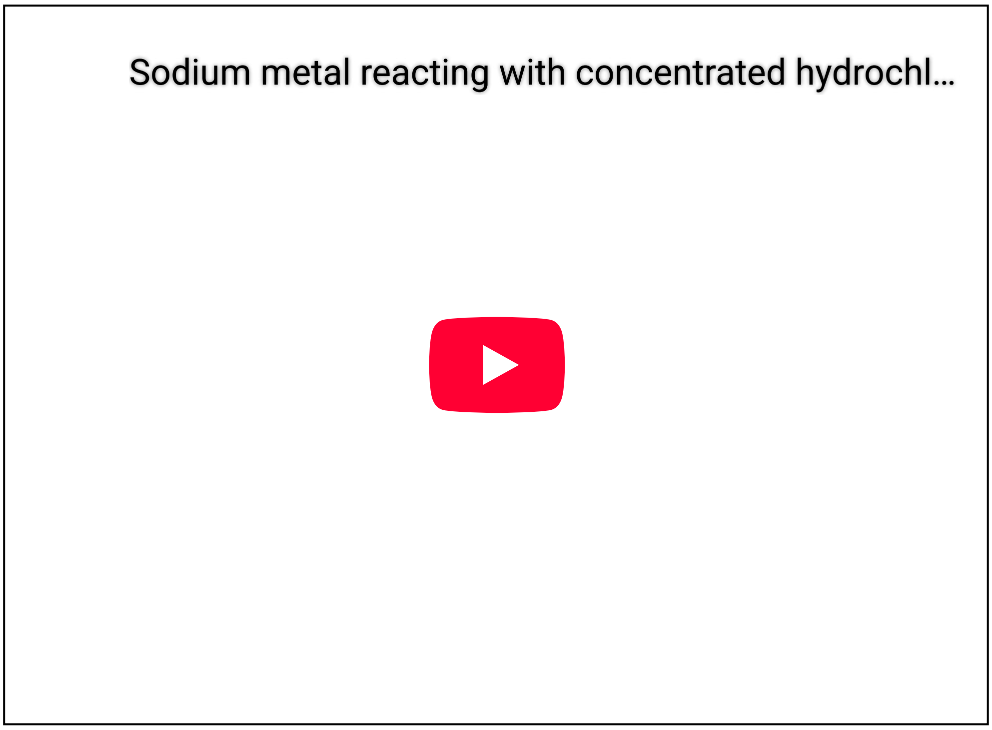
Convert these word equations into balanced chemical equations. You should first complete this on your own and then compare your equations to your partner. Some of the equations will include information that is not required. Figuring out what should be included is key!

1. Zinc and lead(II) nitrate react to produce zinc nitrate and lead.
2. Aluminium bromide and chlorine react to produce aluminium chloride and bromine.
3. Sodium phosphate and calcium chloride will react to yield a precipitate of calcium phosphate and aqueous sodium chloride.
4. Aluminium metal reacts with hydrochloric acid to produce hydrogen gas and aluminium chloride.
5. Dilute hydrochloric acid was added to calcium carbonate powder. Bubbles of carbon dioxide, water and calcium chloride solution were formed. Watch **Video 1** to help you create the balanced chemical equation for this reaction.



Video 1. Reaction of calcium carbonate and hydrochloric acid.

6. A reaction between calcium hydroxide and phosphoric acid produces a solution composed of calcium phosphate and water.
7. In a reaction between copper metal and sulfuric acid, a blue solution was formed composed of copper(II) sulfate, water and bubbles of sulfur dioxide.
8. Hydrogen and nitrogen monoxide, two colourless gases, react together and yield bubbles of nitrogen gas and water.
9. A reaction occurs between sodium and hydrochloric acid to produce bubbles of hydrogen and sodium chloride. Watch **Video 2** to help you create the balanced chemical equation for this reaction.



Video 2. Reaction of sodium and hydrochloric acid.

More information for video 2

1
00:00:00,534 --> 00:00:02,236
[background noise]
2
00:00:02,302 --> 00:00:05,906
Instructor: Here's the reaction of sodium
with concentrated hydrochloric acid,
3
00:00:06,406 --> 00:00:08,008
12 molar hydrochloric acid.
4
00:00:09,209 --> 00:00:12,045
Sodium metal,
I have it sitting on the counter

5
00:00:13,247 --> 00:00:17,117
and I'm going to add
five milliliters of 12 molar
6
00:00:17,384 --> 00:00:21,455
concentrated hydrochloric
acid to the beaker.
7
00:00:21,989 --> 00:00:25,359
Now I'm going to cut
a fresh piece of sodium.
8
00:00:29,763 --> 00:00:32,199
It's a very soft metal, cuts very easily.
9
00:00:40,674 --> 00:00:44,578
I'm going to add the sodium metal
to the hydrochloric acid.